

Cloud Security Envisioning & Strategy

Lab Exercises - Structured Lab Set

Courseware Purpose

This lab set gives instructors and students a structured, scenario-driven path for cloud security planning, posture review, workload protection selection, AI security review, attack path remediation, and 30-day execution planning.

Document	Details
Course code	CESE323 ERC1.0
Document title	CESE323 Lab Exercises - Structured Lab Set
Audience	Students, instructors, facilitators, technical leads, and cloud security stakeholders
Delivery format	Instructor-led workshop, live online delivery, or self-paced scenario-based learning
Prepared for	Skunkworks Academy courseware development
Generated	2026-05-25

Source alignment: Cloud Security Envisioning & Strategy Workshop Syllabus, Microsoft Defender for Cloud posture management, workload protection, AI security, and operational remediation themes.

1. Lab Set Overview

The CESE323 lab set is designed as a practical companion to the Cloud Security Envisioning & Strategy workshop. It moves students from executive security priorities into structured posture review, workload protection selection, AI security assessment, attack path planning, and a 30-day remediation roadmap.

Workshop-to-Lab Alignment

Workshop Theme	Lab Alignment	Expected Student Output
Strategic alignment and governance	Lab 01 maps business outcomes to security priorities.	Security Priority Map
Unified visibility and CSPM	Lab 02 reviews cloud security posture using Defender for Cloud concepts.	Posture Review Checklist
Workload hardening	Lab 03 selects appropriate Defender workload protection plans.	Workload Protection Selection Matrix
AI-specific security	Lab 04 evaluates identity, data, and prompt risks for AI workloads.	AI Security Risk Register
Threat reduction and attack paths	Lab 05 turns findings into attack path remediation.	Attack Path and Remediation Plan
Operational readiness	Lab 06 creates a 30-day execution roadmap.	30-Day Remediation Roadmap

Recommended Lab Flow

Lab	Title	Duration	Mode
01	Security Priority Map	20-30 min	Group or whiteboard
02	Cloud Security Posture Review Checklist	30-45 min	Portal review or guided demo
03	Workload Protection Selection Matrix	25-35 min	Matrix-based analysis
04	AI Security Risk Review	25-35 min	Scenario review
05	Attack Path and Remediation Planning	35-45 min	Attack path workshop
06	30-Day Remediation Roadmap	30-40 min	Capstone planning exercise
Total	Full lab set	165-230 min	Depends on live tenant, demo environment, or paper scenario

Lab Delivery Models

Model	Use Case	Notes
Live Azure tenant	Best for hands-on Defender for Cloud, posture, and workload review.	Requires pre-approved permissions, cost controls, and cleanup.
Instructor demo tenant	Best for live online training where student permissions are limited.	Instructor demonstrates while students complete analysis templates.
Scenario pack	Best for self-paced or restricted environments.	Use screenshots, sample findings, and fictional customer data.
Customer live review	Best for executive envisioning workshops.	Requires written approval, defined scope, and no unapproved production changes.

2. Lab Safety and Operating Guardrails

Important

Do not enable paid Defender plans, deploy agents/extensions, modify production workloads, upload sensitive data, or enter confidential information into AI prompts unless the customer and budget owner have explicitly approved the action in writing.

Minimum Guardrails

- Use named student accounts; avoid shared administrator accounts.
- Use least privilege. Reserve Owner or Global Administrator access for controlled instructor or customer-admin use only.
- Confirm subscription scope before enabling Defender plans or changing policy assignments.
- Use dedicated lab resource groups and disposable test resources where possible.
- Do not use real customer, personal, financial, legal, HR, or production data in lab exercises.
- Define a cleanup plan before creating resources or enabling services that may incur cost.
- Capture configuration changes and evidence in the student workbook or submission template.
- For AI labs, prohibit confidential prompt input and review endpoint exposure, role assignments, and monitoring.

Recommended Permission Model

Role	Recommended Use	Notes
Reader	Observe posture, recommendations, inventory, and compliance.	Suitable for most review-based labs.
Contributor	Create lab resources and perform selected remediation.	Use only in dedicated lab resource groups when possible.
Security Admin	Manage security policies and selected Defender plan settings.	Use for controlled lab leads or instructor demonstrations.
Owner	Enable all required plan capabilities or handle full-scope lab configuration.	Restrict to instructor/customer admin unless explicitly required.

Student Submission Package

- Completed Lab 01 Security Priority Map.
- Completed Lab 02 Cloud Security Posture Review Checklist.
- Completed Lab 03 Workload Protection Selection Matrix.
- Completed Lab 04 AI Security Risk Register.
- Completed Lab 05 Attack Path and Remediation Plan.
- Completed Lab 06 30-Day Remediation Roadmap.
- Supporting screenshots, scenario references, or portal observations where available.
- Final executive summary of no more than one page.

Lab 01: Security Priority Map

Attribute	Details
Estimated duration	20-30 minutes
Primary domain	Strategic security alignment, governance, and risk prioritization
Recommended mode	Group whiteboard or worksheet exercise
Required access	No portal access required; scenario pack or customer context is sufficient

Purpose

Students map business objectives to security priorities so that technical remediation work is connected to executive risk, operational impact, and security governance.

Scenario

A customer has multiple cloud workloads but limited budget and limited engineering capacity. The security team must identify which risks matter most and where Microsoft Defender for Cloud, Microsoft Entra, workload protection, and governance controls should be prioritized first.

Learning Outcomes

- Translate business goals into technical security priorities.
- Identify cloud risk vectors that affect identity, data, workloads, compliance, AI, and security operations.
- Separate executive risk language from low-level implementation detail.
- Produce a short list of priority actions for the remainder of the workshop.

Student Tasks

- 1 Identify the organization's top security drivers, such as compliance, availability, data protection, executive visibility, breach reduction, or AI governance.
- 2 Map each driver to a business impact area such as revenue, reputation, legal exposure, operational continuity, or customer trust.
- 3 Identify the most likely cloud security risks that could undermine each business objective.
- 4 Assign a priority rating of High, Medium, or Low based on business impact and likelihood.
- 5 Recommend the first three security focus areas for the customer or lab scenario.

Security Priority Map Template

Business Goal	Security Risk	Impact	Priority	Recommended Action
Protect customer data	Sensitive storage data exposed or not classified	Regulatory and reputational impact	High	Review storage controls, classify data, and validate Defender coverage
Reduce breach likelihood	Over-privileged identities and weak access controls	Privilege escalation and lateral movement	High	Review privileged roles, MFA, Conditional Access, and least privilege
Improve board visibility	No consolidated posture view or risk trend	Limited governance and delayed remediation	Medium	Use Defender for Cloud Secure Score, recommendations, and workbooks

Business Goal	Security Risk	Impact	Priority	Recommended Action
Adopt AI safely	Prompt/data leakage and unmanaged AI access	Confidentiality and compliance exposure	Medium	Create AI asset register and review AI identity and data controls

Evidence and Submission Requirements

- Completed priority map with at least four business goals and four mapped risks.
- Clearly stated top three security focus areas.
- Short rationale explaining why each priority was selected.
- One executive summary paragraph suitable for a customer stakeholder.

Assessment Rubric

Criteria	Marks	Evidence of Competence
Business-risk alignment	3	Risks are connected to business outcomes, not just technology issues.
Priority logic	3	High, Medium, and Low ratings are defensible and based on impact and likelihood.
Recommended actions	2	Actions are practical, specific, and aligned to the workshop scope.
Executive clarity	2	Summary is clear, concise, and suitable for non-technical stakeholders.

Lab 02: Cloud Security Posture Review Checklist

Attribute	Details
Estimated duration	30-45 minutes
Primary domain	CSPM, Secure Score, recommendations, compliance, inventory, and posture management
Recommended mode	Guided portal review or instructor-provided screenshots
Required access	Reader or higher on the subscription/resource group; Defender for Cloud must be visible

Purpose

Students perform a structured cloud security posture review using Microsoft Defender for Cloud concepts such as Secure Score, recommendations, regulatory compliance, inventory, workbooks, and posture findings.

Scenario

The customer wants to understand their current security posture before enabling additional Defender workload protection plans. Students must review posture signals and convert observations into prioritized findings.

Learning Outcomes

- Assess Defender for Cloud coverage and visibility.
- Identify posture weaknesses, misconfigurations, and compliance gaps.
- Capture high-value recommendations and distinguish quick wins from strategic remediation.
- Produce a concise posture review suitable for technical remediation planning.

Student Tasks

- 1 Confirm the review scope: subscriptions, management groups, resource groups, and critical assets.
- 2 Review Secure Score and identify the lowest-scoring control areas.
- 3 Review recommendations and capture at least five findings.
- 4 Review regulatory compliance status and note missing standards or control gaps.
- 5 Review inventory and confirm whether critical workload types are represented.
- 6 Prioritize quick wins versus longer-term remediation items.

Cloud Security Posture Review Checklist

Area	Review Question	Finding	Risk Level	Action Required
Subscription coverage	Is Defender for Cloud enabled and visible?	Coverage confirmed or gap identified	High/Medium/Low	Enable or validate Defender for Cloud scope
Secure Score	Which control areas have the lowest score?	Lowest-scoring controls captured	High/Medium/Low	Prioritize recommendations with high score impact

Area	Review Question	Finding	Risk Level	Action Required
Recommendations	Which recommendations are high impact?	Top recommendations captured	High	Assign owners and target dates
Identity	Are there over-privileged roles?	Privileged access exposure identified	High	Review RBAC, PIM, MFA, and Conditional Access
Compliance	Are regulatory standards assigned?	Standards missing or control gaps visible	Medium	Assign standards and review failed controls
Visibility	Are logs, alerts, workbooks, and inventory available?	Visibility gaps captured	Medium	Enable diagnostic visibility and define reporting cadence

Evidence and Submission Requirements

- Completed checklist with at least five findings.
- Three prioritized remediation actions with owners or accountable teams.
- Screenshots or scenario references showing the source of the findings where available.
- Short explanation of the difference between immediate fixes and strategic posture improvements.

Assessment Rubric

Criteria	Marks	Evidence of Competence
Coverage review	2	Subscription/resource scope and Defender visibility are clearly documented.
Finding quality	3	Findings are specific, observable, and linked to posture or compliance signals.
Prioritization	3	Remediation priorities reflect risk, effort, and business impact.
Operational usefulness	2	Output can be used by an engineering or security operations team.

Lab 03: Workload Protection Selection Matrix

Attribute	Details
Estimated duration	25-35 minutes
Primary domain	Defender plan selection for Storage, Containers, App Service, and AI Services
Recommended mode	Matrix-based analysis using scenario inventory or live inventory
Required access	Reader access is sufficient for planning; Security Admin, Contributor, or Owner may be needed for enablement

Purpose

Students decide which Microsoft Defender workload protection plans should be enabled based on workload type, business criticality, threat exposure, operational readiness, and cost sensitivity.

Scenario

The customer cannot enable every paid Defender plan immediately. Students must recommend a staged workload protection approach based on risk and business value. The workshop supports selection across Defender for Storage, Defender for Containers, Defender for App Service, and Defender for AI Services.

Learning Outcomes

- Match workload types to appropriate Defender workload protection plans.
- Apply a consistent scoring model for business criticality and threat exposure.
- Account for budget, licensing, deployment effort, and operational ownership.
- Recommend the first two workload protection plans to enable.

Student Tasks

- 1 Review the workload inventory or scenario asset list.
- 2 Identify critical workloads and classify them by business impact.
- 3 Match each workload to its most relevant Defender plan.
- 4 Score business criticality and threat exposure on a 1-4 scale.
- 5 Select the first two Defender plans to enable and justify the decision.
- 6 Document cost, permission, deployment, and monitoring considerations.

Workload Protection Selection Matrix

Workload	Example Asset	Business Criticality	Threat Exposure	Recommended Defender Plan	Priority
Storage	Blob storage account	High/Medium/Low	High/Medium/Low	Defender for Storage	1/2/3
Containers	AKS cluster	High/Medium/Low	High/Medium/Low	Defender for Containers	1/2/3
App Service	Public web app	High/Medium/Low	High/Medium/Low	Defender for App Service	1/2/3

Workload	Example Asset	Business Criticality	Threat Exposure	Recommended Defender Plan	Priority
AI Services	Azure AI or Microsoft Foundry workload	High/Medium/Low	High/Medium/Low	Defender for AI Services	1/2/3

Evidence and Submission Requirements

- Completed matrix for all four workload categories.
- Business criticality and threat exposure score for each workload.
- Final recommendation for the first two Defender plans to enable.
- Documented assumptions about cost, permissions, and operational monitoring.

Assessment Rubric

Criteria	Marks	Evidence of Competence
Plan selection accuracy	3	Recommended Defender plans match the workload types and risks.
Scoring consistency	2	Criticality and exposure are scored using the same logic throughout.
Decision rationale	3	The first two plans are justified by risk, impact, and feasibility.
Cost and operations awareness	2	Budget, permission, and monitoring implications are included.

Lab 04: AI Security Risk Review

Attribute	Details
Estimated duration	25-35 minutes
Primary domain	AI workload risk, identity, data exposure, prompt injection, monitoring, and governance
Recommended mode	Scenario review, AI asset review, or guided Microsoft Foundry/Azure AI discussion
Required access	Scenario only, or Azure AI/Foundry access for live review

Purpose

Students review AI workload risks, including identity control, data exposure, prompt injection, model misuse, endpoint exposure, monitoring, and governance maturity.

Scenario

The customer is adopting Azure AI or Microsoft Foundry workloads. Security leadership needs an initial AI security review before expanding AI usage across business units.

Learning Outcomes

- Identify AI workload assets, owners, users, and data flows.
- Identify AI-specific risks that traditional cloud controls may not fully address.
- Map AI risks to controls across Microsoft Defender for Cloud, Microsoft Entra, Microsoft Purview, and platform governance.
- Create a risk register that supports safe AI adoption.

Student Tasks

- 1 Identify the AI workload, use case, owner, users, and business purpose.
- 2 Identify the data handled by the AI system, including prompts, context, files, outputs, logs, and embeddings where applicable.
- 3 Review who can deploy, invoke, configure, or monitor the AI workload.
- 4 Identify possible threats such as prompt injection, data leakage, unsafe output, overly permissive endpoints, unmanaged model access, and missing logging.
- 5 Map each risk to a recommended control.
- 6 Document residual risk and escalation requirements.

AI Security Risk Register

Risk Area	Example Question	Risk Rating	Recommended Control	Owner
Identity	Who can deploy, invoke, or manage the AI workload?	High/Medium/Low	RBAC, least privilege, managed identities, access reviews	Security/Identity
Data exposure	Is sensitive data entered into prompts or outputs?	High	DLP, data classification, prompt controls, data handling rules	Data Owner

Risk Area	Example Question	Risk Rating	Recommended Control	Owner
Prompt injection	Could users manipulate model behavior or system instructions?	Medium/High	Guardrails, input validation, monitoring, safe prompt design	AI Platform
Model access	Are endpoints public or overly permissive?	Medium/High	Private endpoints, network restrictions, authentication controls	Cloud Platform
Logging	Are AI interactions monitored and retained appropriately?	Medium	Defender alerts, audit logs, data and AI security dashboard	SOC
Governance	Are AI workloads inventoried and owned?	Medium	AI asset register, policy assignment, approval workflow	Risk/Governance

Evidence and Submission Requirements

- AI Security Risk Register containing at least five risks.
- Risk rating and mitigation recommendation for each risk.
- Clear data handling rule for prompts, files, logs, and outputs.
- Statement of residual risk and escalation path for high-risk AI workloads.

Assessment Rubric

Criteria	Marks	Evidence of Competence
AI-specific risk identification	3	Risks include AI-specific threat patterns, not only generic cloud risks.
Identity and data control mapping	3	Controls address both access and data exposure.
Governance completeness	2	Ownership, data handling, and monitoring are included.
Practicality	2	Recommendations can be implemented or validated within the customer environment.

Lab 05: Attack Path and Remediation Planning

Attribute	Details
Estimated duration	35-45 minutes
Primary domain	Attack path analysis, exposure management, least privilege, and remediation planning
Recommended mode	Guided scenario, Microsoft Defender for Cloud attack path view, or whiteboard exercise
Required access	Reader for observation; elevated permissions may be required for remediation

Purpose

Students analyze likely cloud attack paths and produce a remediation plan that reduces blast radius, removes choke points, and improves security posture.

Scenario

A posture review has identified over-privileged identities, exposed resources, and misconfigured workloads. Students must convert those findings into a practical attack path remediation plan.

Learning Outcomes

- Identify how multiple weak controls combine into an attack path.
- Distinguish initial access, privilege escalation, lateral movement, data access, and persistence risks.
- Map technical remediations to attack path steps.
- Assign ownership and effort for prioritized remediation.

Student Tasks

- 1 Select one high-risk finding or scenario attack path.
- 2 Identify the affected asset, identity, network exposure, and data or workload impact.
- 3 Describe the possible attacker path from initial access through impact.
- 4 Identify the enabling weaknesses at each step.
- 5 Recommend remediation actions that break the attack chain.
- 6 Assign ownership, effort, target date, and priority.

Attack Path and Remediation Plan

Attack Path Step	Weakness	Potential Impact	Remediation	Owner	Priority
Initial access	Public endpoint or weak identity control	Unauthorized access to workload	Restrict access, enforce MFA, apply Conditional Access	Identity/Cloud	High
Privilege escalation	Over-privileged role assignment	Expansion of attacker privileges	Reduce RBAC scope, use PIM, remove stale assignments	Identity	High
Lateral movement	Poor segmentation or broad network access	Movement across workloads	Apply segmentation, NSG rules, private endpoints	Network	Medium

Attack Path Step	Weakness	Potential Impact	Remediation	Owner	Priority
Data access	Sensitive storage exposed	Data exfiltration or leakage	Harden storage access, classify data, enable monitoring	Data Owner	High
Persistence	Weak monitoring or alerting	Delayed detection and response	Enable alerts, review incidents, configure SOC workflow	SOC	Medium

Evidence and Submission Requirements

- One completed attack path with at least four attack steps.
- At least one remediation action for each step.
- Ownership, priority, and effort documented.
- Short explanation of which remediation most effectively breaks the attack path.

Assessment Rubric

Criteria	Marks	Evidence of Competence
Attack path logic	3	The path is plausible and follows a coherent attacker sequence.
Weakness identification	2	Each step identifies a specific enabling weakness.
Remediation quality	3	Remediations directly reduce exposure or break the attack chain.
Ownership and prioritization	2	Plan includes realistic owners, priorities, and execution considerations.

Lab 06: 30-Day Remediation Roadmap

Attribute	Details
Estimated duration	30-40 minutes
Primary domain	Execution planning, roadmap development, ownership, and measurable security improvement
Recommended mode	Capstone exercise using outputs from Labs 01-05
Required access	No portal access required; uses findings gathered during previous labs

Purpose

Students convert findings from the previous labs into a realistic 30-day remediation roadmap that balances risk reduction, cost, effort, and business disruption.

Scenario

The customer needs a practical security improvement plan that can start immediately after the workshop. The roadmap must be understandable by executives and actionable by technical teams.

Learning Outcomes

- Consolidate findings from strategic, posture, workload, AI, and attack path reviews.
- Prioritize remediation into immediate, tactical, and operational phases.
- Define owners, success measures, and governance cadence.
- Create an executive-ready 30-day security roadmap.

Student Tasks

- 1 Consolidate the most important findings from Labs 01-05.
- 2 Prioritize the top five remediation actions.
- 3 Categorize work into quick wins, tactical fixes, and strategic improvements.
- 4 Assign owners or accountable teams.
- 5 Define measurable outcomes and success criteria.
- 6 Prepare an executive summary that explains expected risk reduction.

30-Day Remediation Roadmap

Timeframe	Focus Area	Action	Owner	Success Measure
Days 1-7	Immediate risk reduction	Disable risky access, review privileged roles, confirm Defender coverage	Identity/Cloud	Critical exposure reduced and Defender visibility confirmed
Days 8-14	CSPM and compliance	Address high-priority recommendations and compliance gaps	Security/GRC	Secure Score and compliance status improved

Timeframe	Focus Area	Action	Owner	Success Measure
Days 15-21	Workload protection	Enable selected Defender plans and verify coverage	Cloud Platform	Priority workloads protected and coverage workbook validated
Days 22-30	Operational readiness	Configure reporting, alert review, ownership, and governance cadence	SOC/Governance	Repeatable review and remediation process established

Evidence and Submission Requirements

- Roadmap containing at least five remediation actions.
- Each action has an owner, timeframe, risk rationale, and success measure.
- Executive summary describing expected risk reduction.
- Clear distinction between immediate controls and longer-term improvements.

Assessment Rubric

Criteria	Marks	Evidence of Competence
Consolidation of findings	2	Roadmap reflects outputs from earlier labs.
Prioritization and sequencing	3	Actions are ordered logically by risk, dependency, and effort.
Measurability	2	Success measures are specific and observable.
Executive readiness	3	Summary is suitable for leadership and supports decision-making.

Appendix A: Consolidated Student Deliverable Checklist

Deliverable	Minimum Completion Standard	Complete?
Security Priority Map	At least four business goals, mapped risks, priorities, and actions.	Yes / No
Posture Review Checklist	At least five findings and three prioritized remediation actions.	Yes / No
Workload Protection Selection Matrix	All four workload categories scored with first two plans recommended.	Yes / No
AI Security Risk Register	At least five AI-related risks with controls and owners.	Yes / No
Attack Path Remediation Plan	At least one realistic attack path with remediation for each step.	Yes / No
30-Day Remediation Roadmap	Top five actions sequenced with owner, timeframe, and success measure.	Yes / No
Executive Summary	Clear business-level summary of expected risk reduction.	Yes / No

Appendix B: Instructor Marking Summary

Lab	Marks	Main Evidence
Lab 01 - Security Priority Map	10	Business-risk mapping, prioritization, recommended actions, executive clarity
Lab 02 - Cloud Security Posture Review Checklist	10	Coverage, findings, prioritization, operational usefulness
Lab 03 - Workload Protection Selection Matrix	10	Plan selection, scoring, rationale, cost and operations awareness
Lab 04 - AI Security Risk Review	10	AI-specific risk identification, identity/data controls, governance, practicality
Lab 05 - Attack Path and Remediation Planning	10	Attack path logic, weakness identification, remediation quality, ownership
Lab 06 - 30-Day Remediation Roadmap	10	Consolidation, sequencing, measurability, executive readiness
Total	60	Recommended conversion to course percentage as required by delivery model

Appendix C: Instructor Facilitation Notes

- Use Lab 01 to anchor the course in business outcomes before students enter portal-level detail.
- Use Lab 02 to establish the evidence base for the remaining labs.
- Use Lab 03 to force prioritization instead of enabling every available paid plan by default.
- Use Lab 04 to elevate AI as a security, identity, data, and governance concern rather than only an application development concern.
- Use Lab 05 to show how individual misconfigurations combine into exploitable paths.
- Use Lab 06 as the capstone, requiring students to present a practical remediation roadmap.

References and Source Alignment

Reference	How It Informs This Lab Set
Cloud Security Envisioning & Strategy Workshop Syllabus	Defines strategic alignment, unified visibility, threat reduction, workload hardening, operational readiness, and hands-on/live environment review themes.
Microsoft Learn collection for CESE323	Supports the workshop's Microsoft Defender for Cloud, Microsoft Entra, AI security, CSPM, workload protection, and remediation learning paths.
Student pre-course lab readiness requirements	Informs permissions, tenant readiness, safety warnings, budget controls, and live environment guardrails.